Derrick Quinn

Ithaca, USA
dq55@cornell.edu • derrickquinn.github.io

Research

Research area: ML Algorithms and Architectures.

Vision: I co-design novel algorithms and near-data processing architectures to accelerate dense retrieval and long-context inference. My long-term goal is to develop generalized and scalable architectures for Neural Memory Systems, expanding their capability, adaptability, and sustainability across diverse computing environments.

Education

Cornell University — **Ithaca, NY**
Ph.D. Electrical and Computer Engineering — *2024–05/29 (expected)*
Advisor: Mohammad Alian

University of Kansas — **Lawrence, KS**
B.S. Computer Science — *2018–2024*

University of Kansas — **Lawrence, KS**
B.S. Mathematics — *2018–2024*

Awards and Honors

MLCommons
ML and Systems Rising Star — *2026*

Selected Publications

EDBT (2027), accepted — **Lille, France**
ORCHID: Bringing Hybrid Vector Search to IVF [6]
Derrick Quinn, Martin Prammer, Yaochen Li, Kevin Skadron, José F. Martínez, Jignesh M. Patel, Mohammad Alian

IEEE/ACM MICRO (2026) — **Athens, Greece**
C3: Collaborative CPU–Commodity DRAM Computation for Reliable LLM Acceleration [5]
Harry Ji, Emine Ezgi Yücel, Nicole Li, P. Ramakrishna, Derrick Quinn, Jungmin Choi, Hoshik Kim, Mohammad Alian, José F. Martínez

IEEE/ACM MICRO (2025) — **Seoul, South Korea**
LongSight: Compute-Enabled Memory to Accelerate Large-Context LLMs via Sparse Attention [4]
Derrick Quinn, Emine Ezgi Yücel, Jinkwon Kim, José F. Martínez, Mohammad Alian

ACM/IEEE ISCA (2025) — **Tokyo, Japan**
DReX: Accurate and Scalable Dense Retrieval Acceleration via Algorithmic-Hardware Codesign [3]
Derrick Quinn*, Emine Ezgi Yücel*, Martin Prammer, Zhenxing Fan, Kevin Skadron, Jignesh Patel, José F. Martínez, Mohammad Alian

ACM ASPLOS (2025) — **Rotterdam, Netherlands**
Accelerating Retrieval-Augmented Generation [2]
Derrick Quinn, Mohammad Nouri, Neel Patel, John Salihu, Alireza Salemi, Sukhan Lee, Hamed Zamani, Mohammad Alian

IEEE/ACM MICRO (2023) — **Toronto, Canada**
XFM: Accelerated Software-Defined Far Memory [1]
Neel Patel, Amin Mamandipoor, Derrick Quinn, Mohammad Alian

* Equal contribution.

Professional Experience

SEMRON — **Austin, TX**
Systems Architecture Research Intern — *Jan. 2026–July 2026*
- Led development of a unified full-stack simulation framework for SEMRON's capacitive in-memory-computing architecture, integrating functional and hardware models to support architecture exploration, compiler development, and cross-team evaluation.
- Drove cross-layer system-architecture research spanning hardware–software interfaces, ISA design, memory hierarchy and 3D integration, and LLM mapping and pipelining to guide scalable, energy-efficient inference.

Department of Electrical and Computer Engineering, Cornell University — **Ithaca, USA**
Research Assistant — *2024–Current*

Department of Electrical Engineering and Computer Science, University of Kansas — **Lawrence, KS**
Research Assistant — *2022–2024*

Kansas Algebra Program; Department of Mathematics, University of Kansas — **Lawrence, KS**
Algebra Tutor and Class Assistant — *2021–2022*

Dell Technologies — **Overland Park, KS**
Engineering Intern — *2020*